

Project: LifeTrack - Clinical Trials & Patient Engagement Management Platform

1. Introduction

LifeTrack is a web-based Clinical Trials and Patient Engagement Management Platform designed for pharmaceutical companies, CROs (Contract Research Organizations), and healthcare providers to streamline trial operations, patient enrollment, and compliance monitoring. The platform consolidates protocol data, patient records, site activities, and regulatory documentation to provide end-to-end visibility, exception handling, and proactive notifications.

It supports RESTful APIs with Angular/React frontends, and backend implementations in Java Spring Boot or ASP.NET Core. The architecture emphasizes secure handling of patient data, compliance with GxP and HIPAA, and scalability for multi-site trials.

Actors / Users

- **Clinical Trial Manager:** Oversees trial progress, site performance, and compliance.
- **Investigator / Physician:** Manages patient enrollment, visits, and adverse event reporting.
- **Patient / Participant:** Views trial schedule, submits feedback, and receives reminders.
- **Regulatory Officer:** Reviews documentation, compliance reports, and audit logs.
- **Data Manager:** Manages trial data, monitors quality, and prepares reports.
- **Admin:** Configures protocols, sites, users, and integrations.

2. Module Overview

2.1 Identity & Access Management

Authentication, RBAC, audit trails, and compliance logging.

2.2 Protocol & Site Management

Trial protocol registry, site setup, investigator assignments.

2.3 Patient Enrollment & Visit Tracking

Enrollment workflows, visit scheduling, and patient engagement.

2.4 Adverse Event & Exception Management

Capture adverse events, deviations, and resolution workflows.

2.5 Document & Compliance Management

Regulatory document repository, version control, and audit trails.

2.6 Analytics & Reporting

Trial KPIs, enrollment trends, site performance, and compliance dashboards.

2.7 Notifications & Alerts

In-app alerts, email/SMS reminders, and escalation workflows.

3. Architecture Overview

- **Frontend:** Angular or React SPA for dashboards and workflows.
- **Backend:** RESTful microservices (Spring Boot / ASP.NET Core).
- **Database:** Relational DB (Postgres/MySQL/SQL Server) for trial data and patient records.
- **Eventing:** Kafka or cloud pub/sub for enrollment and adverse event notifications.
- **Cache:** Redis for session state and hot lookups.
- **Integration:** EHR/EMR systems via HL7/FHIR; regulatory portals via secure APIs.
- **Security:** TLS, encryption at rest, RBAC, immutable audit logs, HIPAA/GxP compliance.

4. Module Wise Design

4.1 Identity & Access Management Module

Features

- User provisioning, RBAC, secure sessions, audit logs.

Entities

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)

4.2 Protocol & Site Management Module

Features

- Protocol registry, site setup, investigator assignments.

Entities

- **Protocol**(ProtocolID, Title, Phase, StartDate, EndDate, Status)
- **Site**(SiteID, Name, Location, InvestigatorID, Status)

4.3 Patient Enrollment & Visit Tracking Module

Features

- Enrollment workflows, visit scheduling, reminders.

Entities

- **Patient**(PatientID, Name, DOB, ContactInfo, EnrollmentStatus)
- **Visit**(VisitID, PatientID, ProtocolID, VisitDate, Status, Notes)

4.4 Adverse Event & Exception Management Module

Features

- Capture adverse events, deviations, resolution workflows.

Entities

- **AdverseEvent**(EventID, PatientID, ProtocolID, Description, Severity, Status, ReportedDate)
- **Deviation**(DeviationID, ProtocolID, SiteID, Description, Severity, Status)

4.5 Document & Compliance Management Module

Features

- Document repository, version control, audit trails.

Entities

- **Document**(DocumentID, ProtocolID, Type, Version, UploadedBy, UploadedAt, Status)

4.6 Analytics & Reporting Module

Features

- KPIs, enrollment trends, site performance, compliance dashboards.

Entities

- **KPIReport**(ReportID, Scope (Protocol/Site/Period), Metrics (EnrollmentRate, DropoutRate, AECCount), GeneratedDate)

4.7 Notifications & Alerts Module

Features

- Alerts for visits, adverse events, compliance deadlines.

Entities

- **Notification**(NotificationID, UserID, Message, Category (Visit/AE/Compliance), Status, CreatedDate)

5. Deployment Strategy

- **Local/Dev:** Angular/React UI, Spring Boot/.NET backend, local DB.
- **Staging:** Containerized services with synthetic trial data.
- **Production:** Kubernetes orchestration, API gateway, managed RDBMS with replicas, Kafka for events, Redis for caching.
- **High Availability:** Multi-AZ deployments, auto-scaling, disaster recovery playbooks.

6. Database Design

Tables

- **User**(UserID, Name, Role, Email, Phone)

- **AuditLog**(AuditID, UserID, Action, Timestamp)
- **Protocol**(ProtocolID, Title, Phase, StartDate, EndDate, Status)
- **Site**(SiteID, Name, Location, InvestigatorID, Status)
- **Patient**(PatientID, Name, DOB, ContactInfo, EnrollmentStatus)
- **Visit**(VisitID, PatientID, ProtocolID, VisitDate, Status, Notes)
- **AdverseEvent**(EventID, PatientID, ProtocolID, Description, Severity, Status, ReportedDate)
- **Deviation**(DeviationID, ProtocolID, SiteID, Description, Severity, Status)
- **Document**(DocumentID, ProtocolID, Type, Version, UploadedBy, UploadedAt, Status)
- **KPIReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Trial Manager Dashboard:** Protocols, site performance, enrollment KPIs.
- **Investigator Console:** Patient enrollment, visit schedule, adverse event reporting.
- **Patient Portal:** Trial schedule, reminders, feedback submission.
- **Compliance Dashboard:** Document repository, audit logs, regulatory deadlines.
- **Admin Console:** Protocol setup, site configuration, user/role management.

8. Non-Functional Requirements

- **Performance:** Handle up to **20,000 concurrent users**; visit scheduling latency under **300 ms**.
- **Security:** RBAC, TLS, encryption at rest, HIPAA/GxP compliance, immutable audit logs.
- **Scalability:** Horizontal scaling for enrollment and reporting services; multi-tenant support.
- **Availability:** Target **99.9% uptime** for enrollment and compliance services.
- **Maintainability:** Modular APIs, versioning, automated migrations, CI/CD pipelines.
- **Observability:** Centralized logs, metrics, tracing, and alerting.
- **Compliance:** Configurable retention policies, audit trails, regulatory reporting formats.

9. Assumptions & Constraints

- **Phase 1 scope:** Protocol registry, site setup, patient enrollment, visit tracking, adverse event capture, and dashboards.

- **Excluded in Phase 1:** Advanced patient engagement features (mobile apps, wearables), AI-driven trial optimization.
- **Data availability:** Historical trial data and patient records available for reporting and calibration.
- **Integration:** EHR/EMR systems via HL7/FHIR; regulatory portals via secure APIs.
- **Implementable:** Using Java Spring Boot or ASP.NET Core, Angular/React, Kafka, Redis, and relational DB tooling.